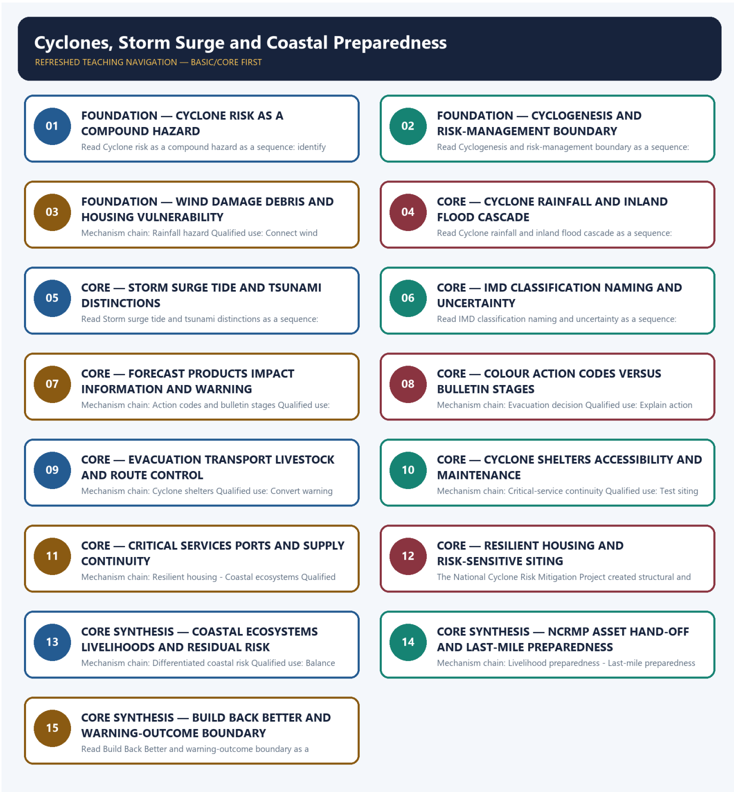
SOLVED PRACTICE WORKBOOK

Cyclones, Storm Surge and Coastal Preparedness - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Cyclones, Storm Surge and Coastal Preparedness - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Compound cyclone hazard?

A. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. B. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation.

Answer: A. Explanation: Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q2. Which option preserves the risk or institutional boundary of Compound cyclone hazard?

A. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. B. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.

Answer: B. Explanation: Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q3. Which statement uses Compound cyclone hazard without changing its hazard, mandate or status?

A. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. B. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. C. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: C. Explanation: Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q4. Which option avoids the standard UPSC close-option trap about Compound cyclone hazard?

A. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. B. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. C. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. D. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.

Answer: D. Explanation: Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q5. Which statement correctly identifies Cyclogenesis boundary?

A. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. B. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point.

Answer: A. Explanation: Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q6. Which option preserves the risk or institutional boundary of Cyclogenesis boundary?

A. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. B. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. C. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. D. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness.

Answer: B. Explanation: Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q7. Which statement uses Cyclogenesis boundary without changing its hazard, mandate or status?

A. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. B. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. C. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: C. Explanation: Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q8. Which option avoids the standard UPSC close-option trap about Cyclogenesis boundary?

A. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. B. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. C. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. D. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation.

Answer: D. Explanation: Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q9. Which statement correctly identifies Wind hazard?

A. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. B. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. C. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. D. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.

Answer: A. Explanation: Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q10. Which option preserves the risk or institutional boundary of Wind hazard?

A. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. B. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. C. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. D. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.

Answer: B. Explanation: Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q11. Which statement uses Wind hazard without changing its hazard, mandate or status?

A. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. B. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. C. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: C. Explanation: Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q12. Which option avoids the standard UPSC close-option trap about Wind hazard?

A. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. B. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. C. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. D. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point.

Answer: D. Explanation: Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q13. Which statement correctly identifies Rainfall hazard?

A. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. B. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. C. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: A. Explanation: Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q14. Which option preserves the risk or institutional boundary of Rainfall hazard?

A. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. B. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. C. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. D. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.

Answer: B. Explanation: Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q15. Which statement uses Rainfall hazard without changing its hazard, mandate or status?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard.

Answer: C. Explanation: Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q16. Which option avoids the standard UPSC close-option trap about Rainfall hazard?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. D. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness.

Answer: D. Explanation: Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q17. Which statement correctly identifies Storm surge?

A. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. B. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. C. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: A. Explanation: Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q18. Which option preserves the risk or institutional boundary of Storm surge?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. C. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. D. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.

Answer: B. Explanation: Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q19. Which statement uses Storm surge without changing its hazard, mandate or status?

A. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. D. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.

Answer: C. Explanation: Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q20. Which option avoids the standard UPSC close-option trap about Storm surge?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.

Answer: D. Explanation: Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q21. Which statement correctly identifies Surge tide tsunami firewall?

A. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. B. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. C. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. D. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard.

Answer: A. Explanation: Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q22. Which option preserves the risk or institutional boundary of Surge tide tsunami firewall?

A. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. B. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. C. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. D. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.

Answer: B. Explanation: Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q23. Which statement uses Surge tide tsunami firewall without changing its hazard, mandate or status?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. C. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. D. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.

Answer: C. Explanation: Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q24. Which option avoids the standard UPSC close-option trap about Surge tide tsunami firewall?

A. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. B. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Answer: D. Explanation: Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q25. Which statement correctly identifies IMD classification?

A. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. B. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. C. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. D. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.

Answer: A. Explanation: IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q26. Which option preserves the risk or institutional boundary of IMD classification?

A. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. B. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. C. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. D. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.

Answer: B. Explanation: IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q27. Which statement uses IMD classification without changing its hazard, mandate or status?

A. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. B. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. C. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. D. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.

Answer: C. Explanation: IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q28. Which option avoids the standard UPSC close-option trap about IMD classification?

A. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. B. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard.

Answer: D. Explanation: IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q29. Which statement correctly identifies Forecast and warning?

A. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.

Answer: A. Explanation: IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q30. Which option preserves the risk or institutional boundary of Forecast and warning?

A. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. B. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.

Answer: B. Explanation: IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q31. Which statement uses Forecast and warning without changing its hazard, mandate or status?

A. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. B. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. C. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. D. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.

Answer: C. Explanation: IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q32. Which option avoids the standard UPSC close-option trap about Forecast and warning?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. C. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. D. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.

Answer: D. Explanation: IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q33. Which statement correctly identifies Action codes and bulletin stages?

A. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. D. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.

Answer: A. Explanation: Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. The other options change the hazard or risk category, institutional owner, legal instrument,

warning stage, evidence rung or implementation status.

Q34. Which option preserves the risk or institutional boundary of Action codes and bulletin stages?

A. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. B. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.

Answer: B. Explanation: Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q35. Which statement uses Action codes and bulletin stages without changing its hazard, mandate or status?

A. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. B. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. C. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. D. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.

Answer: C. Explanation: Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q36. Which option avoids the standard UPSC close-option trap about Action codes and bulletin stages?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.

Answer: D. Explanation: Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q37. Which statement correctly identifies Evacuation decision?

A. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. B. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.

Answer: A. Explanation: Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q38. Which option preserves the risk or institutional boundary of Evacuation decision?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.

Answer: B. Explanation: Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q39. Which statement uses Evacuation decision without changing its hazard, mandate or status?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. C. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. D. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence.

Answer: C. Explanation: Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q40. Which option avoids the standard UPSC close-option trap about Evacuation decision?

A. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. B. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. C. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. D. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.

Answer: D. Explanation: Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to

self-evacuate. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q41. Which statement correctly identifies Cyclone shelters?

A. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. B. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.

Answer: A. Explanation: Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q42. Which option preserves the risk or institutional boundary of Cyclone shelters?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. C. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. D. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.

Answer: B. Explanation: Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q43. Which statement uses Cyclone shelters without changing its hazard, mandate or status?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. C. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. D. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.

Answer: C. Explanation: Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q44. Which option avoids the standard UPSC close-option trap about Cyclone shelters?

A. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. B. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. C. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. D. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.

Answer: D. Explanation: Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q45. Which statement correctly identifies Critical-service continuity?

A. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. B. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. C. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. D. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.

Answer: A. Explanation: Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q46. Which option preserves the risk or institutional boundary of Critical-service continuity?

A. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. B. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. C. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. D. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure.

Answer: B. Explanation: Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q47. Which statement uses Critical-service continuity without changing its hazard, mandate or status?

A. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. B. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. C. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. D. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information.

Answer: C. Explanation: Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. The other options

change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q48. Which option avoids the standard UPSC close-option trap about Critical-service continuity?

A. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. B. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.

Answer: D. Explanation: Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q49. Which statement correctly identifies Resilient housing?

A. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. B. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. C. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. D. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.

Answer: A. Explanation: Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q50. Which option preserves the risk or institutional boundary of Resilient housing?

A. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. B. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence.

Answer: B. Explanation: Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q51. Which statement uses Resilient housing without changing its hazard, mandate or status?

A. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. B. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. C. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. D. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.

Answer: C. Explanation: Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering

details. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q52. Which option avoids the standard UPSC close-option trap about Resilient housing?

A. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.

Answer: D. Explanation: Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q53. Which statement correctly identifies Coastal ecosystems?

A. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. B. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence.

Answer: A. Explanation: Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q54. Which option preserves the risk or institutional boundary of Coastal ecosystems?

A. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. B. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.

Answer: B. Explanation: Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q55. Which statement uses Coastal ecosystems without changing its hazard, mandate or status?

A. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. D. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability.

Answer: C. Explanation: Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q56. Which option avoids the standard UPSC close-option trap about Coastal ecosystems?

A. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure.

Answer: D. Explanation: Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q57. Which statement correctly identifies NCRMP status?

A. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. D. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information.

Answer: A. Explanation: The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q58. Which option preserves the risk or institutional boundary of NCRMP status?

A. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. B. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.

Answer: B. Explanation: The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q59. Which statement uses NCRMP status without changing its hazard, mandate or status?

A. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. D. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Answer: C. Explanation: The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q60. Which option avoids the standard UPSC close-option trap about NCRMP status?

A. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. B. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence.

Answer: D. Explanation: The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q61. Which statement correctly identifies Differentiated coastal risk?

A. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. B. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.

Answer: A. Explanation: Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q62. Which option preserves the risk or institutional boundary of Differentiated coastal risk?

A. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. B. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Answer: B. Explanation: Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. The other

options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q63. Which statement uses Differentiated coastal risk without changing its hazard, mandate or status?

A. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. B. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. C. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. D. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Answer: C. Explanation: Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q64. Which option avoids the standard UPSC close-option trap about Differentiated coastal risk?

A. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. B. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. C. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. D. Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.

Answer: D. Explanation: Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q65. Which statement correctly identifies Livelihood preparedness?

A. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. D. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability.

Answer: A. Explanation: Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q66. Which option preserves the risk or institutional boundary of Livelihood preparedness?

A. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. B. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.

Answer: B. Explanation: Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q67. Which statement uses Livelihood preparedness without changing its hazard, mandate or status?

A. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. B. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. C. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. D. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation.

Answer: C. Explanation: Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q68. Which option avoids the standard UPSC close-option trap about Livelihood preparedness?

A. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. B. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. C. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. D. Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information.

Answer: D. Explanation: Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q69. Which statement correctly identifies Last-mile preparedness?

A. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. B. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. C. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. D. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability.

Answer: A. Explanation: Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q70. Which option preserves the risk or institutional boundary of Last-mile preparedness?

A. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. B. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. C. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. D. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation.

Answer: B. Explanation: Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q71. Which statement uses Last-mile preparedness without changing its hazard, mandate or status?

A. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. B. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. C. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. D. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point.

Answer: C. Explanation: Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q72. Which option avoids the standard UPSC close-option trap about Last-mile preparedness?

A. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. B. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. C. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. D. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.

Answer: D. Explanation: Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q73. Which statement correctly identifies Build Back Better?

A. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. B. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. C. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. D. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.

Answer: A. Explanation: Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q74. Which option preserves the risk or institutional boundary of Build Back Better?

A. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. B. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. C. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. D. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.

Answer: B. Explanation: Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q75. Which statement uses Build Back Better without changing its hazard, mandate or status?

A. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. B. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. C. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. D. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point.

Answer: C. Explanation: Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q76. Which option avoids the standard UPSC close-option trap about Build Back Better?

A. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. B. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability.

Answer: D. Explanation: Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q77. Which statement correctly identifies Warning-outcome firewall?

A. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. B. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation. C. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. D. Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.

Answer: A. Explanation: A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q78. Which option preserves the risk or institutional boundary of Warning-outcome firewall?

A. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. B. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. Warm water, atmospheric instability, moisture, Coriolis force, a pre-existing disturbance and limited vertical wind shear support tropical cyclogenesis, but disaster management focuses on the resulting risk and action chain rather than detailed meteorological derivation.

Answer: B. Explanation: A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q79. Which statement uses Warning-outcome firewall without changing its hazard, mandate or status?

A. Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. B. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. C. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. D. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.

Answer: C. Explanation: A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q80. Which option avoids the standard UPSC close-option trap about Warning-outcome firewall?

A. Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. B. Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. C. Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. D. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Answer: D. Explanation: A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The 2022 GS-I warning-colour card is directly routed. The two 2024 GS-III cards are conservative resilience and flood-cascade applications and retain Topics 01 and 08 as primary owners.

PYQ DEMAND CARD 1 - 2022 GS-I

Demand: Discuss the meaning of colour-coded weather warnings for cyclone-prone areas.

Status: Verified direct routing: Discuss the meaning · 10 marks · 150 words; the solution must distinguish action colours from the separate cyclone bulletin sequence.

Model solution: Forecast and warning: IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Action codes and bulletin stages:** Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. **Evacuation decision:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Last-mile preparedness:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Warning-outcome firewall:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2024 GS-III

Demand: Describe the elements that determine disaster resilience.

Status: Verified direct ownership remains Topic 01; this conservative card routes shelters, lifeline continuity, resilient housing, ecosystems, livelihoods and recovery as cyclone-resilience elements.

Model solution: Cyclone shelters: Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. **Critical-service continuity:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Resilient housing:** Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. **Coastal ecosystems:** Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. **Livelihood preparedness:** Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. **Last-mile preparedness:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Build Back Better:** Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. **Warning-outcome firewall:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2024 GS-III

Demand: Discuss causes, cases, policies and frameworks for urban flooding.

Status: Verified direct ownership remains Topic 08; this card is limited to cyclone rainfall, storm surge and service-cascade contributions to coastal and urban flooding.

Model solution: Compound cyclone hazard: Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. **Rainfall hazard:** Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. **Storm surge:** Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. **Forecast and warning:** IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Evacuation decision:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Critical-service continuity:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Warning-outcome firewall:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish cyclone wind, rainfall and storm-surge hazards. Answer in about 150 words.

Model thesis: Claim: Compound cyclone hazard. **Named evidence/example:** Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wind hazard. **Named evidence/example:** Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rainfall hazard. **Named evidence/example:** Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Storm surge. **Named evidence/example:** Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Surge tide tsunami firewall. **Named evidence/example:** Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.
- Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point.
- Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness.
- Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide.
- Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing.

Qualified conclusion: Claim: Compound cyclone hazard. **Named evidence/example:** Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Wind hazard. **Named evidence/example:** Cyclone wind can damage roofs, weak structures, trees, transmission systems and communications, while debris and prolonged service interruption extend impacts beyond the landfall point. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Rainfall hazard. **Named evidence/example:** Cyclone rainfall can produce riverine, flash, pluvial and urban flooding well inland, so coastal landfall warnings must connect to catchment and city preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Storm surge. **Named evidence/example:** Storm surge is abnormal coastal water-level rise driven chiefly by cyclone winds and low pressure; impact varies with storm track, intensity, coast shape, bathymetry and tide. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Surge tide tsunami firewall. **Named evidence/example:** Storm surge is not an astronomical tide and not a tsunami; tide can modify the total coastal water level, while tsunami generation involves sudden water displacement rather than cyclone forcing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain how IMD classification, forecasts, action codes and bulletin stages differ. Answer in about 150 words.

Model thesis: Claim: IMD classification. **Named evidence/example:** IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Action codes and bulletin stages. **Named evidence/example:** Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard.
- IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.
- Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.

Qualified conclusion: Claim: IMD classification. **Named evidence/example:** IMD classifies cyclonic disturbances by maximum sustained wind speed and names systems from the Cyclonic Storm stage; exact category thresholds should be quoted only from the current official IMD standard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Action codes and bulletin stages. **Named evidence/example:** Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse the warning-to-evacuation and cyclone-shelter preparedness chain. Answer in about 250 words.

Model thesis: Claim: Forecast and warning. **Named evidence/example:** IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Action codes and bulletin stages. **Named evidence/example:** Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation decision. **Named evidence/example:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cyclone shelters. **Named evidence/example:** Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome.
- Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence.
- Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.
- Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.
- Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.
- A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Qualified conclusion: Claim: Forecast and warning. **Named evidence/example:** IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Action codes and bulletin stages. **Named evidence/example:** Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation decision. **Named evidence/example:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cyclone shelters. **Named evidence/example:** Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine resilient housing, critical services and coastal ecosystems as a layered cyclone-risk portfolio. Answer in about 250 words.

Model thesis: Claim: Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Resilient housing. **Named evidence/example:** Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal ecosystems. **Named evidence/example:** Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Differentiated coastal risk. **Named evidence/example:** Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.
- Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.
- Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure.
- Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.

Qualified conclusion: Claim: Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Resilient housing. **Named evidence/example:** Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal ecosystems. **Named evidence/example:** Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Differentiated coastal risk. **Named evidence/example:** Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's coastal cyclone preparedness after NCRMP, focusing on maintenance, last-mile action and differentiated risk. Answer in about 300 words.

Model thesis: Claim: Evacuation decision. **Named evidence/example:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cyclone shelters. **Named evidence/example:** Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NCRMP status. **Named evidence/example:** The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Differentiated coastal risk. **Named evidence/example:** Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Livelihood preparedness. **Named evidence/example:** Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate.
- Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness.
- Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.
- The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence.

- Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free.
- Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information.
- Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.
- A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Qualified conclusion: Claim: Evacuation decision. **Named evidence/example:** Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Cyclone shelters. **Named evidence/example:** Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** NCRMP status. **Named evidence/example:** The National Cyclone Risk Mitigation Project created structural and non-structural coastal-risk assets through completed phases; asset operation, maintenance and current readiness now require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Differentiated coastal risk. **Named evidence/example:** Cyclone frequency, coast geometry, exposure, housing, poverty, ecosystems and local capacity differ across coasts, so a lower-category or less-frequent area is not risk-free. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Livelihood preparedness. **Named evidence/example:** Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a Build Back Better framework for cyclone-affected coastal communities and livelihoods. Answer in about 300 words.

Model thesis: Claim: Compound cyclone hazard. **Named evidence/example:** Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Resilient housing. **Named evidence/example:** Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal ecosystems. **Named evidence/example:** Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Livelihood preparedness. **Named evidence/example:** Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Build Back Better. **Named evidence/example:** Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services.
- Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans.
- Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details.
- Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure.

- Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information.
- Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action.
- Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability.
- A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Qualified conclusion: Claim: Compound cyclone hazard. **Named evidence/example:** Tropical-cyclone risk combines destructive wind, intense rainfall, storm surge, waves, river or urban flooding, erosion and cascading failure of power, communications, transport and health services. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Critical-service continuity. **Named evidence/example:** Hospitals, emergency operations, power, water, telecom, roads, ports and supply chains need redundancy, shutdown protocols, rapid assessment and restoration plans. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Resilient housing. **Named evidence/example:** Risk-sensitive siting, code-compliant construction, roof and connection safety, maintenance and safer repair reduce housing vulnerability; exam answers should not prescribe engineering details. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal ecosystems. **Named evidence/example:** Mangroves, dunes, wetlands and other coastal ecosystems can moderate some wind-wave-surge effects and support livelihoods, but they complement rather than replace warnings, shelters and resilient infrastructure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Livelihood preparedness. **Named evidence/example:** Fishers, farmers, coastal workers, vendors and tourism-dependent households need vessel and gear safety, market and income continuity, livestock arrangements and timely reopening information. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Last-mile preparedness. **Named evidence/example:** Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Build Back Better. **Named evidence/example:** Recovery should restore housing, services, ecosystems and livelihoods with safer siting and construction, risk-informed finance and social protection rather than recreate pre-cyclone vulnerability. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.